
Handwriting Author Identification



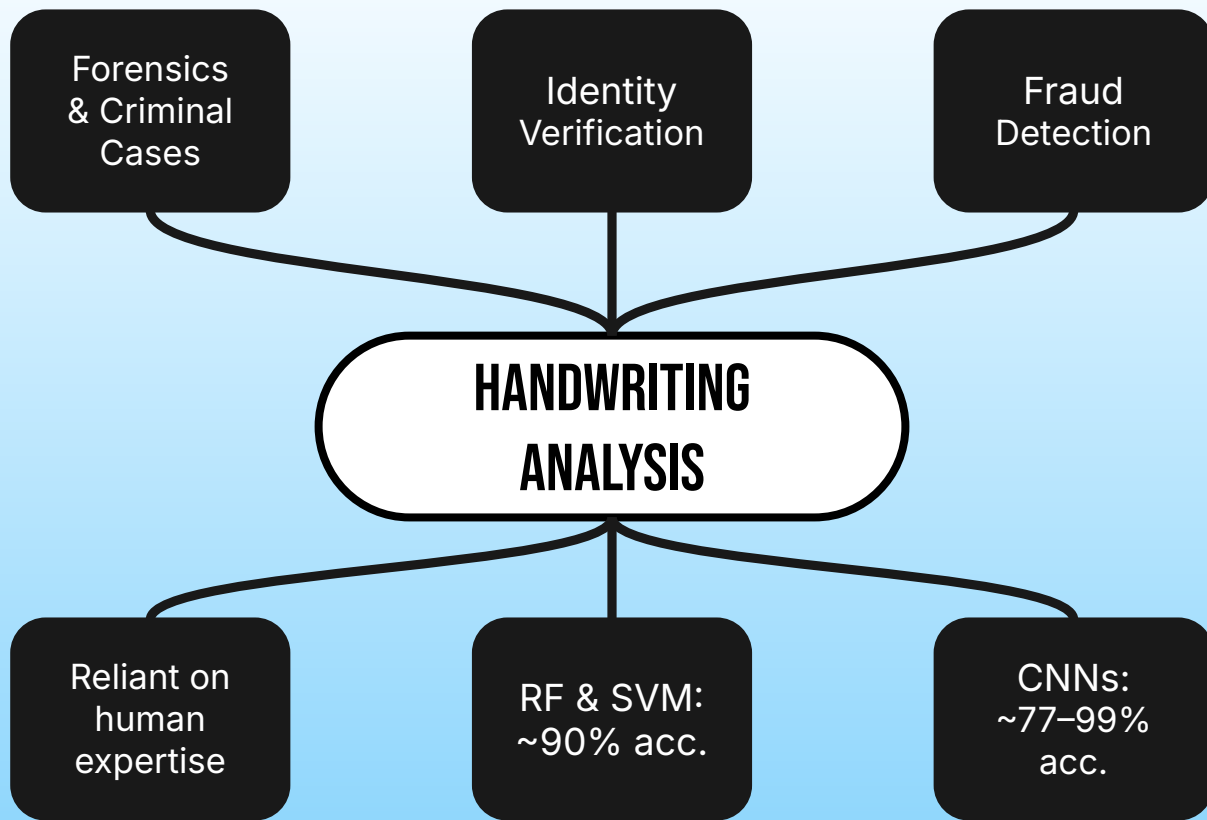
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Data Capstone I



01

Handwriting Analysis

Why do we need it?



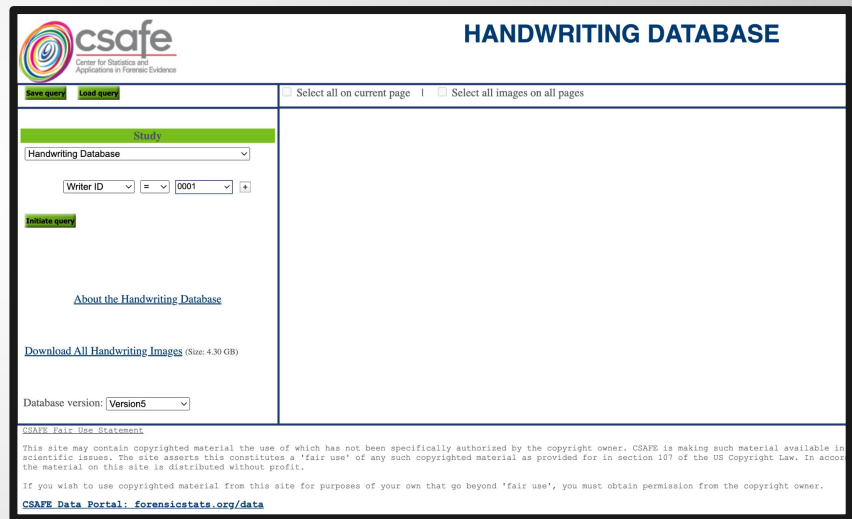


Dataset



→ CSAFE's Handwriting Database

- ◆ CSAFE: Center for Statistics and Applications in Forensic Evidence
 - Handwriting Database, version 5
- ◆ Image data
 - Grayscale
 - ~2521x2810
- ◆ 475 classes x 27 images per class = **12,825 instances**
- ◆ CSV file with writer info



The screenshot shows the CSAFE Handwriting Database web interface. At the top, the CSAFE logo and name are on the left, and the title 'HANDWRITING DATABASE' is on the right. Below the header, there are two buttons: 'Save query' and 'Load query'. To the right of these buttons are two checkboxes: 'Select all on current page' and 'Select all images on all pages'. The main content area is divided into two columns. The left column has a green header 'Study' and contains a dropdown menu set to 'Handwriting Database'. Below this is a 'Writer ID' field with a dropdown set to '0001' and a '+' button. There is a 'Initiate query' button. Below the query section are links for 'About the Handwriting Database' and 'Download All Handwriting Images (Size: 4.30 GB)'. At the bottom of the left column is a 'Database version' dropdown set to 'Version5'. The right column is empty. At the bottom of the page, there is a 'CSAFE Fair Use Statement' section with text about copyright and a link to the 'CSAFE Data Portal: forensicstate.org/data'.

Our London business is good, but
Vienna and Berlin are quiet..

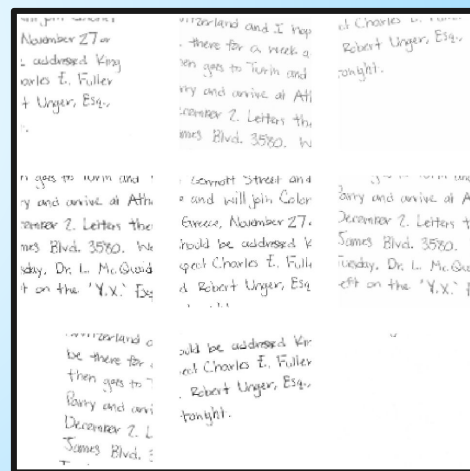
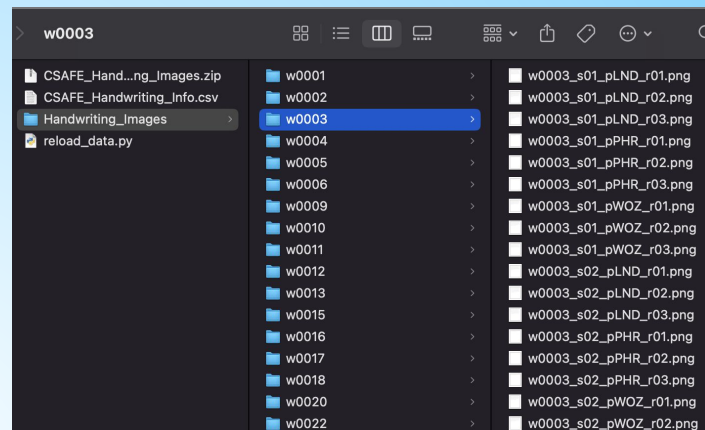
Mr. D. Lloyd has gone to Switzerland
And I hope for good news. He will
be there for a week at 1496 Zornott
Street and then goes to Turin
and Rome and will join Colonel
Parry and arrive at Athens Greece
November 27 or December 2. Letters
there should be addressed
King James Blvd 3580, We expect
Charles F. Fuller Tuesday
Dr. McQuaid and Robert Unger
Esq. left on the 'Y.X.' Express
tonight

Within a short time she was walking briskly
toward the Emerald City, her silver shoes tinkling
merely on the hard, yellow roadbed. The
sun shone bright and the birds sang sweet
and Dorothy did not feel nearly as bad as
you might think a little girl would who had
been suddenly whisked away from her own
country and set down in the midst of a
strange land.

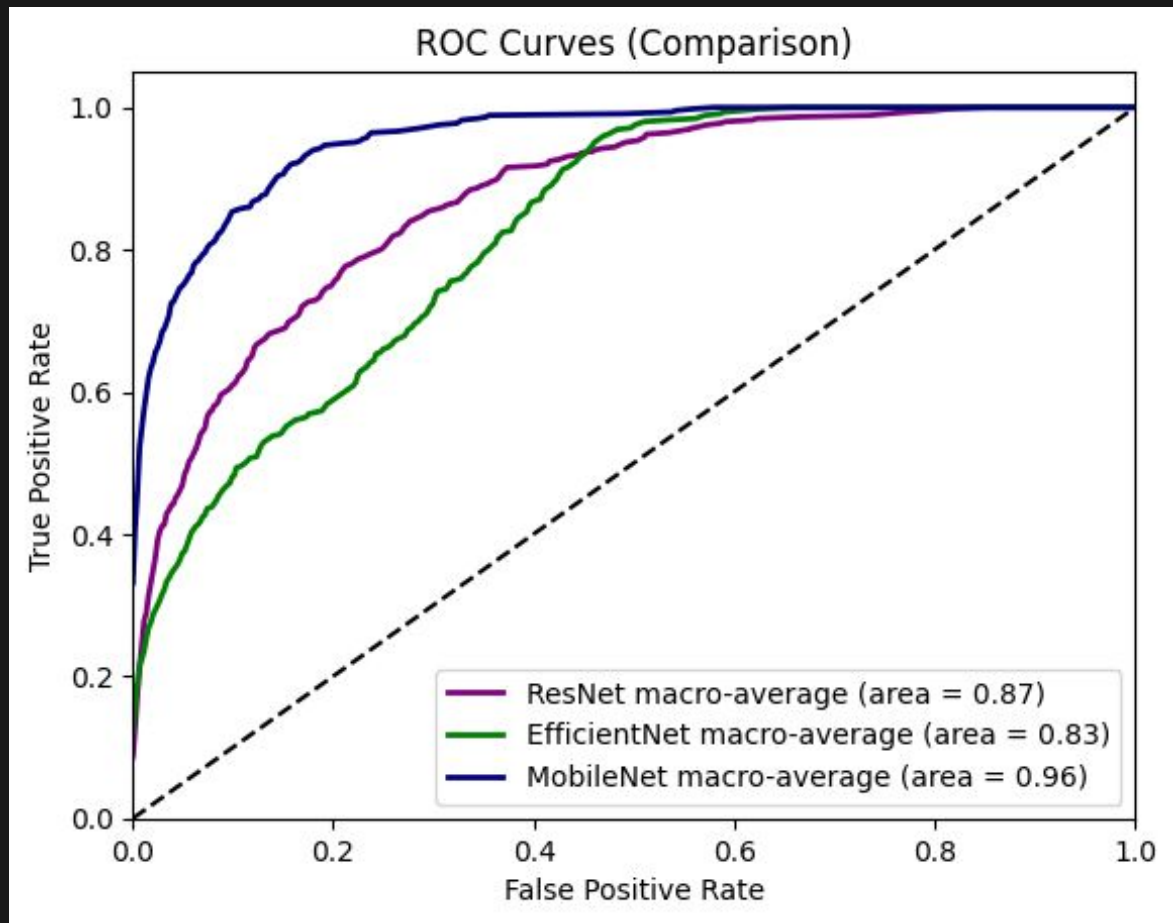
The early bird may get the worm, but
the second mouse gets the cheese.

Methodology

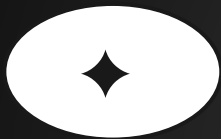
- ◆ **Model(s):** Convolutional Neural Nets (CNNs)
 - ResNet, MobileNet, EfficientNet
- ◆ **Libraries:** keras, tensorflow, scikit-learn
- ◆ **Preprocessing:**
 - Organizing files (train-test; 80/20)
 - Validation?
 - Hyperparameter tuning: batch size, image size, # of epochs, etc.
- ◆ **Metric:** macro-average
- ◆ **Concerns:** overfitting, limited per-class data, generalizability of model, balancing computational feasibility with image quality



Baseline

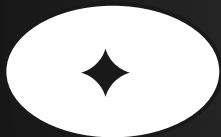


→ Outcomes



Yield a Robust Model

To yield a robust model with code, pipeline, & documentation for reproducibility in a GitHub repo.



Standardize Handwriting Identification

Show the process of identifying authorship from handwriting using ML is feasible, faster, and more objective.



Future Work

Further fine-tune models, try alternative techniques (e.g., Siamese NN approach), reduce & project model outputs into visual space for interpretability.

→ Citations

- ◆ Crawford, A., Ray, A., Carriquiry, A., Kruse, J., & Peterson, M. (2019). CSAFE Handwriting Database. <https://doi.org/10.25380/u002Fiastate.10062203.v1>. Dataset (version 5).
- ◆ https://github.com/zaisid/DATA4380_Vision/



Questions?

